

A

SDC

This appendix describes the SDC¹ format version 1.7. This format is primarily used to specify the timing constraints of a design. It does not contain any commands for any specific tool such as link and compile. It is a text file. It can be handwritten or created by a program and read in by a program. Some SDC commands are applicable to implementation or synthesis only. However all SDC commands are listed here.

SDC syntax is a TCL-based format, that is, all commands follow the TCL syntax. An SDC file contains the SDC version number at the beginning of the file. This is followed by the design constraints. Optional comments (a comment starts with the # character and ends at the end of the line) can be present in a SDC file interspersed with the design constraints. Long lines in

1. Synopsys Design Constraints. Reproduced here with permission from Synopsys, Inc.

design constraints can be split up across multiple lines using the backslash (\) character.

A.1 Basic Commands

These are the basic commands in SDC.

```
current_instance [instance_pathname]  
# Sets the current instance of design. This allows other  
# commands to set or get attributes from that instance.  
# If no argument is supplied, then the current instance  
# becomes the top-level.
```

Examples:

```
current_instance /core/U2/UPLL  
current_instance ..      # Go up one hierarchy.  
current_instance         # Set to top.
```

```
expr arg1 arg2 . . . argn
```

```
list arg1 arg2 . . . argn
```

```
set variable_name value
```

```
set_hierarchy_separator separator  
# Specifies the default hierarchy separator used within  
# the SDC file. This can be overridden by using the -hsc  
# option in the individual SDC commands where allowed.
```

Examples:

```
set_hierarchy_separator /  
set_hierarchy_separator .
```

```
set_units [-capacitance cap_unit] [-resistance res_units]  
[-time time_unit] [-voltage voltage_unit]  
[-current current_unit] [-power power_unit]
```

Specifies the units used in the SDC file.

Examples:

```
set_units -capacitance pf -time ps
```

A.2 Object Access Commands

These commands specify how to access objects in a design instance.

all_clocks

Returns a collection of all clocks.

Examples:

```
foreach_in_collection clkvar [all_clocks] {  
    . . .  
}  
set_clock_transition 0.150 [all_clocks]
```

all_inputs [-level_sensitive] [-edge_triggered]

[-clock *clock_name*]

Returns a collection of all input ports in the design.

Example:

```
set_input_delay -clock VCLK 0.6 -min [all_inputs]
```

all_outputs [-level_sensitive] [-edge_triggered]

[-clock *clock_name*]

Returns a collection of all output ports in the design.

Example:

```
set_load 0.5 [all_outputs]
```

all_registers [-no_hierarchy] [-clock *clock_name*]

[-rise_clock *clock_name*] [-fall_clock *clock_name*]

[-cells] [-data_pins] [-clock_pins] [-slave_clock_pins]

[-async_pins] [-output_pins] [-level_sensitive]

[-edge_triggered] [-master_slave]

Returns the set of registers with the properties

as specified, if any.

Examples:

all_registers -clock DAC_CLK

Returns all registers clocked by clock DAC_CLK.

current_design [*design_name*]

Returns the name of the current design. If specified with
an argument, it sets the current design to the one
specified.

Examples:

current_design FADD # Sets the current context to FADD.

current_design # Returns the current context.

get_cells [-**hierarchical**] [-**hsc separator**] [-**regexp**]

[-**nocase**] [-**of_objects** *objects*] *patterns*

Returns a collection of cells in the design that match the
specified pattern. Wildcard can be used to match
multiple cells.

Examples:

get_cells RegEdge* # Returns all cells that

match pattern.

foreach_in_collection cvar [**get_cells -hierarchical ***] {

. . .

} # Returns all cells in design by searching

recursively down the hierarchy.

get_clocks [-**regexp**] [-**nocase**] *patterns*

Returns a collection of clocks in the design that match
the specified pattern. When used in context such as -from
or -to, it returns a collection of all flip-flops driven
by the specified clocks.

Examples:

set_propagated_clock [**get_clocks** SYS_CLK]

set_multicycle_path -to [**get_clocks** jtag*]

```
get_lib_cells [-hsc separator] [-regexp] [-nocase]
patterns
# Creates a collection of library cells that are currently
# loaded and those that match the specified pattern.
```

Example:

```
get_lib_cells cmos13lv/AOI3*
```

```
get_lib_pins [-hsc separator] [-regexp] [-nocase]
patterns
# Returns a collection of library cell pins that match the
# specified pattern.
```

```
get_libs [-regexp] [-nocase] patterns
# Returns a collection of libraries that are currently
# loaded in the design.
```

```
get_nets [-hierarchical] [-hsc separator] [-regexp]
[-nocase] [-of_objects objects] patterns
# Returns a collection of nets that match the specified
# pattern.
```

Examples:

```
get_nets -hierarchical *      # Returns list of all nets in
# design by searching recursively down the hierarchy.
get_nets FIFO_patt*
```

```
get_pins [-hierarchical] [-hsc separator] [-regexp]
[-nocase] [-of_objects objects] patterns
# Returns a collection of pin names that match the
# specified pattern.
```

Examples:

```
get_pins *
get_pins U1/U2/U3/UAND/Z
```

```
get_ports [-regexp] [-nocase] patterns
# Returns a collection of port names (inputs and outputs)
# of design that match the specified pattern.
```

Example:

```
foreach_in_collection port_name [get_ports clk*] {  
    # For all ports that start with "clk".  
    . . .  
}
```

Can an object such as a port be referenced without “getting” the object? When there is only one object with that name in the design, there really isn't any difference. However, when multiple objects share the same name, then using the **get_*** commands become more important. It avoids any possible confusion over which type of object is being referred to. Imagine a case where there is a net called *BIST_N1* and a port called *BIST_N1*. Consider the SDC command:

```
set_load 0.05 BIST_N1
```

The question is which *BIST_N1* is being referred to? The net or the port? It is best in most cases to explicitly qualify the type of object, such as in:

```
set_load 0.05 [get_nets BIST_N1]
```

Consider another example of a clock *MCLK* and a port called *MCLK*, and the following SDC command:

```
set_propagated_clock MCLK
```

Does the object refer to the port called *MCLK*, or to the clock called *MCLK*? In this particular case, it refers to the clock since that is what the precedence rules of *set_propagated_clock* command will select. However, to be clear, it is better to qualify the object explicitly, either as:

```
set_propagated_clock [get_clocks MCLK]
```

or as:

```
set_propagated_clock [get_ports MCLK]
```

With this explicit qualification, there is no need to depend on the precedence rules, and the SDC is clear.

A.3 Timing Constraints

This section describes the SDC commands that are related to timing specifications.

```
create_clock -period period_value [-name clock_name]  
  [-waveform edge_list] [-add] [source_objects]  
# Defines a clock.  
# When clock_name is not specified, the clock name is the  
# name of the first source object.  
# The -period option specifies the clock period.  
# The -add option is used to create a clock at a pin that  
# already has an existing clock definition. Else if this  
# option is not used, this clock definition overrides any  
# other existing clock definition at that node.  
# The -waveform option specifies the rising edge and  
# falling edge (duty cycle) of the clock. The default  
# is (0, period/2). If a clock definition is on a path  
# after another clock, then it blocks the previous clock  
# from that point onwards.
```

Examples:

```
create_clock -period 20 -waveform {0 6} -name SYS_CLK \  
  [get_ports SYS_CLK]      # Creates a clock of period  
  # 20ns with rising edge at 0ns and the falling edge  
  # at 6ns.  
create_clock -name CPU_CLK -period 2.33 \  
  -add [get_ports CPU_CLK]  # Adds the clock definition
```

```
# to the port without overriding any existing
# clock definitions.

create_generated_clock [-name clock_name]
  -source master_pin [-edges edge_list]
  [-divide_by factor] [-multiply_by factor]
  [-duty_cycle percent] [-invert]
  [-edge_shift shift_list] [-add] [-master_clock clock]
  [-combinational]
  source_objects
# Defines an internally generated clock.
# If no -name is specified, the clock name is that of the
# first source object.
# The source of the generated clock, specified by -source,
# is a pin or port in the design.
# If more than one clock feeds the source node,
# the -master_clock option must be used to specify which of
# these clocks to use as the source of the generated clock.
# The -divide_by option can be used to specify the clock
# division factor; similarly for -multiply_by.
# The -duty_cycle can be used to specify the new duty cycle
# for clock multiplication.
# The -invert option can be specified if the phase of the
# clock has been inverted.
# Instead of using clock multiplication or division, clock
# generation can also be specified using -edges
# and -edge_shift options. The -edges option specifies a
# list of three numbers specifying the edges of the master
# clock edges to use for the first rising edge, the next
# falling edge, and the next rising edge. For example, a
# clock divider can be specified as -divide_by 2 or
# as -edges {1 3 5}.
# The -edge_shift option can be used in conjunction with
# the -edges option to specify an amount to shift for each
# of the three edges.
Examples:
create_generated_clock -divide_by 2 -source \
```



```
[get_ports sys_clk] -name gen_sys_clk [get_pins UFF/Q]
create_generated_clock -add -invert -edges {1 2 8} \
  -source [get_ports mclk] -name gen_clk_div
create_generated_clock -multiply_by 3 -source \
  [get_ports ref_clk] -master_clock clk10MHz \
  [get_pins UPLL/CLKOUT] -name gen_pll_clk
```

```
group_path [-name group_name] [-default]
  [-weight weight_value] [-from from_list]
  [-rise_from from_list] [-fall_from from_list]
  [-to to_list] [-rise_to to_list] [-fall_to to_list]
  [-through through_list] [-rise_through through_list]
  [-fall_through through_list]
```

Gives a name to the specified group of paths.

```
set_clock_gating_check [-setup setup_value]
  [-hold hold_value] [-rise] [-fall] [-high] [-low]
  [object_list]
```

Provides the ability to specify a clock gating check on
any object.

Clock gating checks are performed only on gates that get
a clock signal.

By default, the setup and hold values are 0.

Examples:

```
set_clock_gating_check -setup 0.15 -hold 0.05 \
  [get_clocks ck20m]
set_clock_gating_check -hold 0.3 \
  [get_cells U0/clk_divider/UAND1]
```

```
set_clock_groups [-name name] [-logically_exclusive]
  [-physically_exclusive] [-asynchronous] [-allow_paths]
  -group clock_list
```

Specifies a group of clocks with the specific
property and assigns a name to the group.

```
set_clock_latency [-rise] [-fall] [-min] [-max]  
  [-source] [-late] [-early] [-clock clock_list] delay  
  object_list
```

Specifies the clock latency for a given clock.
There are two types of latency: network and source.
Source latency is the clock network delay between the
clock definition pin and its source, while *network*
latency is the clock network delay between the clock
definition pin and the flip-flop clock pins.

Examples:

```
set_clock_latency 1.86 [get_clocks clk250]  
set_clock_latency -source -late -rise 2.5 \  
  [get_clocks MCLK]  
set_clock_latency -source -late -fall 2.3 \  
  [get_clocks MCLK]
```

```
set_clock_sense [-positive] [-negative] [-pulse pulse]  
  [-stop_propagation] [-clock clock_list] pin_list
```

Set clock property on pin.

```
set_clock_transition [-rise] [-fall] [-min] [-max]  
  transition clock_list
```

Specifies the clock transition at the clock
definition point.

Examples:

```
set_clock_transition -min 0.5 [get_clocks SERDES_CLK]  
set_clock_transition -max 1.5 [get_clocks SERDES_CLK]
```

```
set_clock_uncertainty [-from from_clock]  
  [-rise_from rise_from_clock]  
  [-fall_from fall_from_clock] [-to to_clock]  
  [-rise_to rise_to_clock] [-fall_to fall_to_clock]  
  [-rise] [-fall] [-setup] [-hold]  
  uncertainty [object_list]
```

Specifies the clock uncertainty for clocks or for
clock-to-clock transfers.

The setup uncertainty is subtracted from the data
 # required time for a path, and the hold uncertainty is
 # added to the data required time for each path.

Examples:

```
set_clock_uncertainty -setup -rise -fall 0.2 \  
  [get_clocks CLK2]  
set_clock_uncertainty -from [get_clocks HSCLK] -to \  
  [get_clocks SYCLK] -hold 0.35
```

```
set_data_check [-from from_object] [-to to_object]  
  [-rise_from from_object] [-fall_from from_object]  
  [-rise_to to_object] [-fall_to to_object]  
  [-setup] [-hold] [-clock clock_object] value
```

Performs the specified check between the two pins.

Example:

```
set_data_check -from [get_pins UBLK/EN] \  
  -to [get_pins UBLK/D] -setup 0.2
```

```
set_disable_timing [-from from_pin_name]  
  [-to to_pin_name] cell_pin_list
```

Disables a timing arc/edge inside the specified cell.

Example:

```
set_disable_timing -from A -to ZN [get_cells U1]
```

```
set_false_path [-setup] [-hold] [-rise] [-fall]  
  [-from from_list] [-to to_list] [-through through_list]  
  [-rise_from rise_from_list] [-rise_to rise_to_list]  
  [-rise_through -rise_through_list]  
  [-fall_from fall_from_list] [-fall_to fall_to_list]  
  [-fall_through fall_through_list]
```

Specifies a path exception that is not to be considered
 # for STA.

Examples:

```
set_false_path -from [get_clocks jtag_clk] \  
  -to [get_clocks sys_clk]  
set_false_path -through U1/A -through U4/ZN
```

```
set_ideal_latency [-rise] [-fall] [-min] [-max]
    delay object_list
# Sets ideal latency to specific objects.
```

```
set_ideal_network [-no_propagate] object_list
# Identifies points in design that are sources of an
# ideal network.
```

```
set_ideal_transition [-rise] [-fall] [-min] [-max]
    transition_time object_list
# Specifies the transition time for the ideal networks
# and ideal nets.
```

```
set_input_delay [-clock clock_name] [-clock_fall]
    [-rise] [-fall] [-max] [-min] [-add_delay]
    [-network_latency_included] [-source_latency_included]
    delay_value port_pin_list
# Specifies the data arrival times at the specified input
# ports relative to the clock specified.
# The default is the rising edge of clock.
# The -add_delay option allows the capability to add more
# than one constraint to that particular pin or port.
# Multiple input delays with respect to different clocks
# can be specified using this -add_delay option.
# By default, the clock source latency of the launch clock
# is added to the input delay value, but when
# the -source_latency_included option is specified, the
# source network latency is not added because it is
# assumed to be factored into the input delay value.
# The -max delay is used for clock setup checks and recovery
# checks, while the -min delay is used for hold and removal
# checks. If only -min or -max or neither is specified,
# the same value is used for both.
```

Examples:

```
set_input_delay -clock SYSCLK 1.1 [get_ports MDIO*]
set_input_delay -clock virtual_mclk 2.5 [all_inputs]
```

```

set_max_delay [-rise] [-fall]
    [-from from_list] [-to to_list] [-through through_list]
    [-rise_from rise_from_list] [-rise_to rise_to_list]
    [-rise_through rise_through_list]
    [-fall_from fall_from_list] [-fall_to fall_to_list]
    [-fall_through fall_through_list]
    delay_value

```

Sets the maximum delay on the specified path.
 # This is used to specify delay between two arbitrary pins
 # instead of from a flip-flop to another flip-flop.

Examples:

```

set_max_delay -from [get_clocks FIFOCLK] \
    -to [get_clocks MAINCLK] 3.5
set_max_delay -from [all_inputs] \
    -to [get_cells UCKDIV/UFF1/D] 2.66

```

```

set_max_time_borrow delay_value object_list
# Sets the max time that can be borrowed when analyzing
# a path to a latch.

```

Example:

```

set_max_time_borrow 0.6 [get_pins CORE/CNT_LATCH/D]

```

```

set_min_delay [-rise] [-fall]
    [-from from_list] [-to to_list] [-through through_list]
    [-rise_from rise_from_list] [-rise_to rise_to_list]
    [-rise_through rise_through_list]
    [-fall_from fall_from_list] [-fall_to fall_to_list]
    [-fall_through fall_through_list]
    delay_value

```

Sets the min delay for the specified path, which can
 # be between any two arbitrary pins.

Examples:

```

set_min_delay -from U1/S -to U2/A 0.6
set_min_delay -from [get_clocks PCLK] \
    -to [get_pins UFF/*/S]

```

```

set_multicycle_path [-setup] [-hold] [-rise] [-fall]
  [-start] [-end] [-from from_list] [-to to_list]
  [-through through_list] [-rise_from rise_from_list]
  [-rise_to rise_to_list]
  [-rise_through rise_through_list]
  [-fall_from fall_from_list] [-fall_to fall_to_list]
  [-fall_through fall_through_list] path_multiplier
# Specifies a path as a multicycle path. Multiple -through
# can also be specified.
# Use the -setup option if the multicycle path is just for
# setup. Use the -hold option if the multicycle path is
# for hold.
# If neither -setup nor -hold is specified, the default
# is -setup and the default hold multiplier is 0.
# The -start refers to the path multiplier being applied to
# the launch clock, while -end refers to the path
# multiplier being applied to the capture clock.
# Default is -start.
# The value of the -hold multiplier represents the number
# of clock edges away from the default hold multicycle
# value which is 0.

```

Examples:

```

set_multicycle_path -start -setup \
  -from [get_clocks PCLK] -to [get_clocks MCLK] 4
set_multicycle_path -hold -from UFF1/Q -to UCNTFF/D 2
set_multicycle_path -setup -to [get_pins UEDGEFF*] 4

```

```

set_output_delay [-clock clock_name] [-clock_fall]
  [-level_sensitive]
  [-rise] [-fall] [-max] [-min] [-add_delay]
  [-network_delay_included] [-source_latency_included]
  delay_value port_pin_list
# Specifies the required time of the output relative
# to the clock. The rising edge is default.
# By default, the clock source latency is added to the
# output delay value but when the -source_latency_included
# option is specified, the clock latency value is not added

```

```
# as it is assumed to be included in the output delay value.
# The -add_delay option can be used to specify multiple
# set_output_delay on a pin/port.
```

```
set_propagated_clock object_list
# Specifies that clock latency needs to be computed,
# that is, it is not ideal.
```

Example:

```
set_propagated_clock [all_clocks]
```

A.4 Environment Commands

This section describes the commands that are used to setup the environment of the design under analysis.

```
set_case_analysis value port_or_pin_list
# Specifies the port or pin that is set to
# the constant value.
```

Examples:

```
set_case_analysis 0 [get_pins UDFT/MODE_SEL]
set_case_analysis 1 [get_ports SCAN_ENABLE]
```

```
set_drive [-rise] [-fall] [-min] [-max]
resistance port_list
# Is used to specify the drive strength of the input port.
# It specifies the external drive resistance to the port.
# A value of 0 signifies highest drive strength.
```

Example:

```
set_drive 0 {CLK RST}
```

```
set_driving_cell [-lib_cell lib_cell_name] [-rise]
[-fall] [-library lib_name] [-pin pin_name]
[-from_pin from_pin_name] [-multiply_by factor]
[-dont_scale] [-no_design_rule]
```

```
[-input_transition_rise rise_time]
[-input_transition_fall fall_time] [-min] [-max]
[-clock clock_name] [-clock_fall] port_list
# Is used to model the drive resistance of the cell
# driving the input port.
Example:
    set_driving_cell -lib_cell BUFX4 -pin ZN [all_inputs]
```

```
set_fanout_load value port_list
# Sets the specified fanout load on the output ports.
Example:
    set_fanout_load 5 [all_outputs]
```

```
set_input_transition [-rise] [-fall] [-min] [-max]
[-clock clock_name] [-clock_fall]
transition port_list
# Specifies the transition time on an input pin.
Examples:
    set_input_transition 0.2 \
        [get_ports SD_DIN*]
    set_input_transition -rise 0.5 \
        [get_ports GPIO*]
```

```
set_load [-min] [-max] [-subtract_pin_load] [-pin_load]
[-wire_load] value objects
# Set the value of capacitive load on pin or net in design.
# The -subtract_pin_load option specifies to subtract the
# pin cap from the indicated load.
Examples:
    set_load 50 [all_outputs]
    set_load 0.1 [get_pins UFF0/Q]      # On an internal pin.
    set_load -subtract_pin_load 0.025 \
        [get_nets UCNT0/NET5]          # On a net.
```

```
set_logic_dc port_list
set_logic_one port_list
```

set_logic_zero *port_list*

Sets the specified ports to be a don't care value,
a logic one or a logic zero.

Examples:

set_logic_dc SE

set_logic_one TEST

set_logic_zero [get_pins USB0/USYNC_FF1/Q]

set_max_area *area_value*

Sets the max area limit for current design.

Example:

set_max_area 20000.0

set_max_capacitance *value object_list*

Specifies the max capacitance for ports or on a design.

If for a design, it specifies the max capacitance for all
pins in the design.

Examples:

set_max_capacitance 0.2 [current_design]

set_max_capacitance 1 [all_outputs]

set_max_fanout *value object_list*

Specifies the max fanout value for ports or on a design.

If for a design, it specifies the max fanout for all
output pins in the design.

Examples:

set_max_fanout 16 [get_pins UDFT0/JTAG/ZN]

set_max_fanout 50 [current_design]

set_max_transition [-clock_path]

[-data_path] [-rise] [-fall] *value object_list*

Specifies the max transition time on a port or

on a design. If for a design, it specifies the max
transition on all pins in a design.

Example:

set_max_transition 0.2 UCLKDIV0/QN

set_min_capacitance *value object_list*
Specifies a minimum capacitance value for a port
or on pins in design.

Example:

```
set_min_capacitance 0.05 UPHY0/UCNTR/B1
```

set_operating_conditions [-**library** *lib_name*]
[-**analysis_type** *type*] [-**max** *max_condition*]
[-**min** *min_condition*] [-**max_library** *max_lib*]
[-**min_library** *min_lib*] [-**object_list** *objects*]
[*condition*]

Sets the specified operating condition for timing
analysis. Analysis type can be *single*, *bc_wc*, or
on_chip_variation. Operating conditions are defined in
libraries using the *operating_conditions* command.

Examples:

```
set_operating_conditions -analysis_type bc_wc  
set_operating_conditions WCCOM  
set_operating_conditions -analysis_type \  
    on_chip_variation
```

set_port_fanout_number *value port_list*
Sets maximum fanout of a port.

Example:

```
set_port_fanout_number 10 [get_ports GPIO*]
```

set_resistance [-**min**] [-**max**] *value list_of_nets*
Sets the resistance on the specified nets.

Examples:

```
set_resistance 10 -min U0/U1/NETA  
set_resistance 50 -max U0/U1/NETA
```

set_timing_derate [-**cell_delay**] [-**cell_check**]
[-**net_delay**] [-**data**] [-**clock**] [-**early**] [-**late**]
derate_value [*object_list*]
Specifies derating values.

set_wire_load_min_block_size *size*

Specifies the minimum block size to be used when the
wire load mode is set to *enclosed*.

Example:

set_wire_load_min_block_size 5000

set_wire_load_mode *mode_name*

Defines the mechanism of how a wire load model is to be
used for nets in a hierarchical design.
The *mode_name* can be *top*, *enclosed*, or *segmented*.
The *top* mode causes the wire load model defined in the
top-level of the hierarchy to be used at all lower levels.
The *enclosed* mode causes the wire load model of the block
that fully encloses that net to be used for that net.
The *segmented* mode causes net segment in the block to use
the block's wire load model.

Example:

set_wire_load_mode enclosed

set_wire_load_model **-name** *model_name* [**-library** *lib_name*]
[-min] [-max] [*object_list*]

Defines the wire load model to be used for the current
design or for the specified nets.

Example:

set_wire_load_model **-name** "eSiliconLightWLM"

set_wire_load_selection_group [**-library** *lib_name*]
[-min] [-max] *group_name* [*object_list*]

Sets the wire load selection group for a design when
determining wire load model based on cell area of the
blocks. Selection groups are typically defined in
technology libraries.

A.5 Multi-Voltage Commands

These commands apply when multi-voltage islands are present in a design.

```
create_voltage_area -name name  
  [-coordinate coordinate_list] [-guard_band_x float]  
  [-guard_band_y float] cell_list
```

```
set_level_shifter_strategy [-rule rule_type]
```

```
set_level_shifter_threshold [-voltage float]  
  [-percent float]
```

```
set_max_dynamic_power power [unit]  
# Specify max dynamic power.
```

Example:

```
set_max_dynamic_power 0 mw
```

```
set_max_leakage_power power [unit]  
# Specify max leakage power.
```

Example:

```
set_max_leakage_power 12 mw
```

□